Navneet Kumar Mishra

ECE Undergraduate | Embedded Systems, Robotics, IoT & VLSI

info.navneet.net@gmail.com | +91 7482037075
Patsa, Hasanpur, Samastipur, Bihar (848206)
LinkedIn | GitHub | Portfolio



Professional Summary

Electronics and Communication Engineering undergraduate (B.Tech, 2028) with a strong foundation in electronics, embedded systems, IoT and **robotics**, and a growing interest in **VLSI** and digital design. Has built complete **ESP32**-based robotic systems end to end, from **SOLIDWORKS** design and **3D printing** to Embedded C firmware and wireless dashboards. Student Coordinator at e-Yantra Robotics Lab and former BSNL intern, seeking internships in electronics, embedded, robotics and VLSI.

Education

B.Tech, Electronics & Communication Engineering — **2024 – 2028 (Expected)**
Government Engineering College Kishanganj | Currently in 5th Semester | **CGPA: 7.56** (up to 3rd Sem)

Intermediate (XII), BSEB — **2024**
Samastipur College, Samastipur | 67.4%

Matriculation (X), CBSE — **2022**
Progressive Central High School, Patsa | 86.4%

Technical Skills

Programming Languages	C, C++, Python
Embedded Systems	Embedded C, Arduino, ESP32, STM32, Raspberry Pi
Communication Protocols	UART, I^2C, SPI, MQTT
Robotics & Automation	ROS2, IoT, Sensor & Actuator Integration, Autonomous Robotics
CAD, Simulation & Design	SOLIDWORKS, MATLAB, 3D Printing, Proteus Design
Development Tools	Git, GitHub, Arduino IDE, STM32CubeIDE, PlatformIO, VS Code
Core Coursework	Electronic Devices & Circuits, Network Theory, Signals & Systems, Object-Oriented Programming
Areas of Interest	Embedded Systems, VLSI & Digital Design, IoT, Communication Systems, Robotics & Automation

Experience & Leadership

Student Coordinator — **Feb 2026 – Present**
e-Yantra Robotics Lab | Government Engineering College Kishanganj

- Coordinate day-to-day robotics lab activities and organise technical sessions for interested students.
- Mentor peers in Embedded Systems, Arduino and Robotics via hands-on experiments and workshops.
- Support student participation in robotics competitions and technical events.

Technical Intern — **Dec 2025 – Jan 2026**
Bharat Sanchar Nigam Limited (BSNL)

- Gained hands-on exposure to telecommunication networks, optical fibre communication, switching and transmission systems.

Projects

IoT Controlled Mobile Robotic Arm — **Feb 2026 – May 2026**
ESP32 | SOLIDWORKS | Embedded C | IoT | ESP32-CAM

- Designed a custom robotic arm in **SOLIDWORKS** and built it with **3D printing** at the e-Yantra Robotics Lab.
- Engineered an ESP32-powered mobile platform integrating sensors and actuators for remote operation.
- Built a responsive web dashboard for wireless control and **real-time system monitoring**.
- Integrated ESP32-CAM for live **video streaming** and remote visual feedback.

IoT Based Metal Detection Robot — **Feb 2026 – May 2026**
ESP32 | Embedded C | Sensors | IoT

- Developed an ESP32-powered robot that detects metal objects and reports them wirelessly.
- Integrated a metal-detection sensor, mobile platform and web interface for **real-time** operation and remote control.
- Implemented instant detection alerts and remote monitoring on a wireless dashboard.
- Optimised the embedded firmware for reliable detection, navigation and **low-power operation**.

Certifications & Achievements

- **e-Yantra Robotics Competition (eYRC) 2026-27** – Participant | IIT Bombay | Ongoing
- **NPTEL (Elite)** – Embedded System Design with ARM | IIT Kharagpur (SWAYAM)
- **e-Yantra eLSI** – Basics of Embedded Systems & Robotics, Certificate of Merit (98/100) | IIT Bombay
- **Cybersecurity Fundamentals** – Tech Mahindra Foundation | Skill India Digital Hub
- **InnovoQuest 2.0** (Innovation & Entrepreneurship Challenge) | **eLSI Sprint** (Embedded Systems & Robotics Challenge)
- **Proteus Design Suite Workshop (Cohort-I)** | GEC Kishanganj | Certificate of Participation, Aug 2026